

Chapter 1 — Dataset management

Keep datasets trustworthy as they are stored, shared, and reused

Learning objectives

- Explain why management matters after models ship
- List core management practices (versioning, metadata, access, catalog)
- Name privacy and security controls that protect sensitive tables

Why management matters

- Quiet work: access, versions, backups, retention
- Traceability for audits and collaboration
- Prevents undocumented preprocessing from poisoning models
- Protects the investment in collection and quality

Best practices (overview)

- Version control for evolving extracts
- Cleaning and transformation with clear schemas
- Metadata documentation and data catalogs
- Storage, backups, and role-based access control

Privacy and security

- Encryption at rest and in transit
- Anonymization / de-identification for sharing
- MFA and role-based access
- Compliance (e.g., GDPR, HIPAA) plus audit logging

Looking ahead

- This chapter: what datasets are and how to use them responsibly
- Next: Chapter 2 — how records are collected
- Management depth continues in later chapters (privacy, docs, storage)

Takeaways

- Management is not an optional add-on
- Versioning + metadata + access must work together
- Protection begins when records are first stored